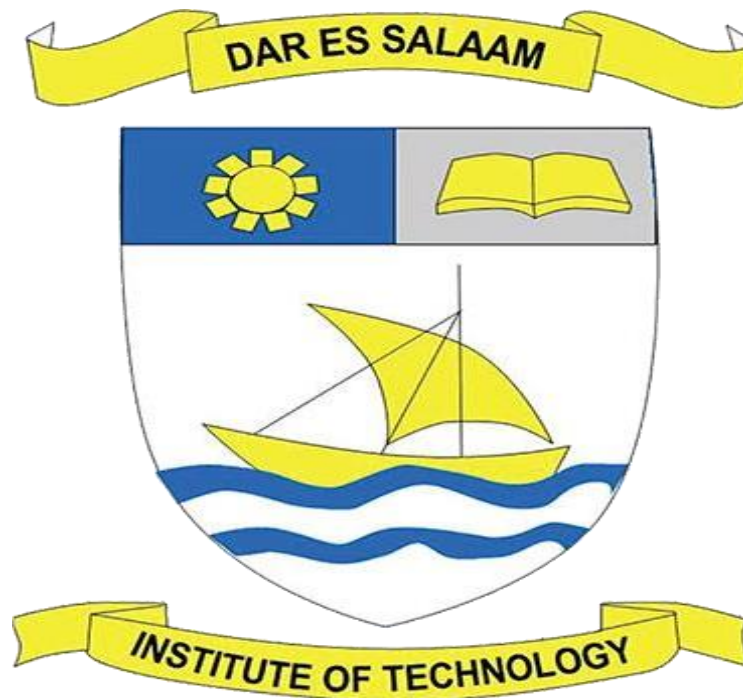


DAR ES SALAAM INSTITUTE OF TECHNOLOGY



DEPARTMENT OF ELECTRICAL ENGINEERING BACHELOR DEGREE IN ELECTRICAL ENGINEERING

NTA LEVEL 8

PROJECT 1

PROJECT TITLE: DESIGN AND IMPLEMENTATION OF INTELLIGENT
AND EFFICIENT TOMATO SORTING MACHINE
WITH PRECISION AT ULANGA FOOD PANTRY

PROJECT TYPE: PROBLEM SOLVING

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ACADEMIC YEAR: 2024/2025

PROJECT SUPERVISOR: ENG. TUMAINI SHIJA

CERTIFICATION

The undersigned certifies has read and hereby recommend for the examination project report titled design and implementation of intelligent and efficient tomato sorting machine with precision at Ulanga food pantry, in fulfillment of the requirements for Bachelor of Engineering in Electrical at Dar es salaam Institute of Technology.

Supervisor's name: TUMAINI SHIJA

Signature:

Date:

DECLARATION

I MORICE KURUSANGA, student of Bachelor degree in Electrical Engineering at Dar es salaam Institute of Technology, would like to declare that the project report entitled design and implementation of intelligent and efficient tomato sorting machine with precision at Ulanga food pantry submitted by me in partial fulfillment of the requirements for the award of Bachelor degree in Electrical Engineering is my original work.

Student's name: MORICE KURUSANGA

Signature:

Date:

ACKNOWLEDGEMENT

I would like to thank the almighty God who provides me with health, strength and reinforces me to understand and undertake this project. I sincerely thank my supervisor **Eng. Tumaini Shija** and project coordinator **Mr. S. Ndola** for their time and effort for guiding us and lending usable advices where necessary and all lectures and colleagues from department of electrical engineering for their advice and great assistance toward the success of this project. “May God bless you all”.

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LIST OF ABBREVIATION

S/N	ACRONYM	MEANING
1.	MCU	Micro controller unit
2.	VDC	Direct Current Voltage
3.	VAC	Alternating Current Voltage
4.	AC	Alternating Current
5.	DC	Direct Current
6.	I/O	Input/output
7.	MHz	Mega Hertz
8.	RGB	Red Green Blue

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CHAPTER ONE

1.0 INTRODUCTION

In the agricultural sector, the sorting and grading of tomatoes play a crucial role in ensuring quality and marketability. Traditional manual sorting methods are often labour-intensive, time-consuming, and prone to human error. To address these challenges, the project aims to develop an automated system that leverages advanced technologies to enhance the sorting process.

This project focuses on creating a sophisticated tomato sorting machine that utilizes sensors, microcontrollers, and machine learning algorithms to accurately classify tomatoes based on their colour, size, and ripeness. By integrating precision and efficiency into the sorting process, the system aims to reduce labour costs, minimize sorting errors, and improve the overall quality of tomatoes distributed by Ulanga Food Pantry.

The intelligent sorting machine will be designed to handle large volumes of tomatoes, ensuring consistent and reliable sorting results. The implementation of this project will not only streamline operations at Ulanga Food Pantry but also set a benchmark for innovative agricultural practices in the region.

In the agricultural sector, the quality and efficiency of post-harvest processes play a crucial role in ensuring that consumers receive fresh and high-quality products. At Ulanga Pantry, the need for an advanced and reliable tomato sorting system has become increasingly evident due to growing demand and the inherent challenges of manual sorting methods.

The intelligent tomato sorting system proposed will focus in reduction of human errors existed in manual sorting of tomatoes, for it will ensure hundred percent of the stored tomatoes in a pantry are those that meet the criteria of being unripe.

The project aims also at reducing cost for effective sorting while ensuring minimal human intervention and incorporating time management. The process has to be precise and within short period of time.

In terms of sustainability, this project has the potential to contribute positively to food security and waste reduction. By sorting and categorizing tomatoes based on quality, the system ensures that only the best tomatoes are processed, and any damaged or low-quality tomatoes can be diverted for alternative uses, such as composting or creating tomato-based products like sauces. This process helps minimize food waste while also enhancing the value of the final product.

The system is divided into three sections namely signal generating unit, controlling unit and storage unit.

For a signal generating unit the system will be made up of two main parts, one the recipient part called hopper where tomatoes will be deployed, second part will consist of a color sensor that will be responsible solely for generating an input signal to the microcontroller for further use based on the color of the tomato.

For the controlling unit, the system will consist of a microcontroller which is the brain of the system, it receives input signal from the sensor, upon receiving it generates an output signal that corresponds to the available input. It will also consist of a motor driver that will control the movement of the robotic arm according to the received signal from the microcontroller. And the last part will be robotic arm that abides to command given by the motor drive, it will rotate to a hopper for picking the tomato and shift it to its respective storage container.

For storage unit, there will be three different storage containers namely unripe container that will store all tomatoes that are not ripe (these will later be taken to the pantry for preservation), ripe container that will store tomatoes which are already ripe (will be for immediate use), and lastly rotten container that will collect all rotten and defected tomatoes through being bitten by birds, affected by pesticides and fertilizers.

By utilizing cutting-edge technologies like machine learning, and automated systems, this project demonstrates the potential of applying modern innovations to traditional industries. The development of such a system can pave the way for future advancements in food processing and automation, encouraging other agricultural businesses to adopt similar practices that enhance both operational efficiency and product quality.

The project also has operational benefits, this project increase impact on the local economy. By improving the productivity and automated tomato sorting at Ulanga Food Pantry, the project could lead to job creation and supply chain of high-quality tomatoes

1.1 BACKGROUND OF THE PROJECT

The recent, tomato sorting, used system involves the use of manual inspection and sorting by the human beings, in which it is not effective in controlling and regulating of sorting and preservation of tomatoes in the pantry due to hindering factors such as fatigue of the personnel.

Traditional tomato sorting methods are often labor-intensive, time-consuming, and prone to human error. The manual sorting process typically involves workers visually inspecting each tomato for defects, size, and color, which can be inefficient and inconsistent. Moreover, human labor in the sorting process can result in physical strain, fatigue, and potential health and safety risks. This reliance on manual labor also increases operational costs, particularly in large-scale operations where significant volumes of tomatoes must be processed quickly to meet demand. Also human workers may have varying levels of skill and experience, leading to inconsistencies in sorting criteria and quality control inconsistencies can impact the overall quality of the sorted tomatoes.

This has led to increasing number of cases where rotten tomatoes are mixed with unripe tomatoes, for it is forbidden to mix rotten and unripe tomatoes during preservation processes since it may lead to accelerated spoilage of unripe ones, quality degradation of preserved tomatoes and leading to a large batch of stored tomatoes to go down as a food waste.

The sorting process can be slow, especially when dealing with large quantities of produce. This can result in delays and reduced efficiency of tomatoes being sorted. As demand for tomatoes grows and production scales goes up, it becomes increasingly difficult to maintain the same level of efficiency with a manual system as it will require an increased number of labor force to sort tomatoes, and it will require a lot of time resources as well as the cost of post-production will increase.

While workers attempt to sort tomatoes accurately, there is always a risk of generating more waste due to human error or oversight. Tomatoes that are not damaged or imperfect may be discarded unnecessarily, resulting in higher levels of waste and reduced number of stored tomatoes at the Ulanga pantry. With many workers involved in sorting, the risk of mishandling tomatoes or causing damage through careless actions increases.

In most cases, manual sorting systems can introduce the risk of contamination if workers are not careful about their hygiene and sanitation. Tomatoes are often handled by many workers; improper handling can lead to contamination, affecting the safety and quality of the final product

1.2 PROBLEM STATEMENT:

The traditional manual sorting methods employed at Ulanga Food Pantry for classifying and sorting tomatoes are labor-intensive, time-consuming, and prone to human error. These methods rely heavily on human labor to visually inspect and sort tomatoes based on various criteria such as color, size, and ripeness. However, this approach presents several challenges, this often leads to inefficient, reduce productivity and increased operational costs. Also human error in sorting can affect the overall quality of the end product, impacting customer satisfaction and market value. This dependency on manual labor makes the sorting process less scalable and efficient, especially during peak harvest seasons but also prolonged periods of manual sorting can lead to physical strain and health issues for workers, impacting their well-being and productivity. Ensuring a safe and ergonomic working environment is essential but challenging in manual sorting operations. Manual sorting is not only time-consuming but also lacks the accuracy needed to identify subtle imperfections, resulting in the misclassification of tomatoes based on size, ripeness, color, and defects. Therefore the reliance on human labor leads to higher costs, increased risk of worker fatigue, and the potential for physical damage to the tomatoes, which negatively impacts the overall yield and quality of the produce

To address these challenges, there is a need for an intelligent and efficient tomato sorting system that leverages advanced technologies such as sensors, microcontrollers, and machine learning algorithms. The proposed solution aims to develop an automated sorting machine that can accurately classify tomatoes based on their color, size, and ripeness, ensuring consistent quality and reducing labor costs. By integrating precision and efficiency into the sorting process, the system will streamline operations at Ulanga Food Pantry, improve the overall quality of distributed tomatoes, and set a benchmark for innovative agricultural practices in the region. This project aims to design and implement an intelligent sorting system that will address the inefficiencies of manual sorting, providing a more accurate, efficient, and sustainable solution for the tomato sorting process. The intelligent sorting system will not only enhance the efficiency and accuracy of the sorting process but also provide a scalable and sustainable solution to meet the growing demands of the community.

1.3 OBJECTIVES

This is divided into two parts, namely: Main objective and Specific objectives

1.3.1 MAIN OBJECTIVE

The main objective of this project is to design and implement of intelligent and efficient tomato sorting machine with precision at ulanga food pantry

1.3.2 SPECIFIC OBJECTIVES

- i. To design an intelligent tomato sorting machine based on machine learning.
- ii. To simulate the assembled circuit.
- iii. To build and test the prototype.

1.4 SIGNIFICANCE OF THE PROJECT

- i. **The proposed system will reduce cost and labor dependency.** Automated systems can work continuously without breaks, fatigue, or variations in performance. This ensures a consistent and reliable sorting process, further reducing the need for manual intervention.
- ii. **The proposed system will ensure consistent quality of sorted tomatoes.** The system uses color sensors to accurately detect the color of each tomato. This ensures that tomatoes are sorted based on their ripeness, with consistent criteria applied to every tomato
- iii. **The proposed system will promote technological advancement in agriculture specifically tomato farming.** The system incorporates advanced sensors and microcontrollers to accurately measure and classify tomatoes based on various parameters such as color, size, and ripeness. This integration of technology enhances the precision and efficiency of the sorting process
- iv. **The proposed system will enable long and efficient preservation of tomatoes.** The system uses advanced sensors to accurately detect the ripeness of each tomato. By sorting tomatoes based on their ripeness, the system ensures that only tomatoes at the desired ripeness level are selected for immediate distribution or storage.

- v. **Minimized Waste and Losses.** Traditional manual sorting processes often result in unnecessary waste, as slightly damaged or under-ripe tomatoes may be discarded without proper evaluation. The intelligent sorting system can accurately identify tomatoes that may not meet the predefined criteria for sorting and kind of tomatoes to be kept.
- vi. **Scalability and Flexibility.** The intelligent sorting system is adaptable and can easily scale to meet increasing demands. As the Ulanga Food Pantry grows or expands its operations, the system can handle larger volumes of tomatoes without compromising performance, making it a future-proof solution for ongoing business development
- vii. **Reduced Labor-Related Risks and Health Concerns.** The manual sorting process is physically demanding and can lead to worker fatigue, repetitive strain injuries, and health issues. By shifting to an automated system, the project mitigates the risks associated with labor-intensive work and improves overall workplace safety.

1.5 METHODOLOGY

This part outlines the specific procedures and techniques used to collect, analyze, and interpret data that will be used to fulfillment of this project. Among the methodology that will be use are as follows;

i. Literature review.

This section consists of an overview of the project as well as information from different sources regarding this project. This section shows the relevance of other literary works to the project. It is the point at which the project is concentrated

ii. Data collection.

A process of gathering and measuring information on targeted variables in an established system, which then enables one to answer relevant questions and evaluate outcomes
Involves site visitation

iii. Data analysis.

A process of inspecting, transforming and modeling data with a goal of forming a useful outcome. Involves systematically examining and interpreting data collected during the project to draw meaningful insights and make informed decisions

iv. Design of the proposed system.

Involves the creation of a plan or specification for the construction of the proposed system.

v. Simulation of the proposed system.

This involves the use of mathematical models to replicate the behavior of the actual circuit.

vi. Circuit simulation.

Provides practical feedback in designing of real-world system hence allows determination of the correctness and efficiency of the design before actual system construction.

vii. Prototype building.

This process includes the transforming of the designed layouts/drawings into a real object.

viii. Prototype testing.

This stage involves the testing the erected system physical so as to see the performance

CHAPTER TWO

2.0 LITERATURE REVIEW

This chapter gives an overview and relevant literature that supports the project work. It introduces various details related to project that are key points of understanding the ways that will be used to achieve the project objectives. It explains about the components, operation and drawback of the existing system. It also explains in detail about the proposed system including components that will be used, operation, Block diagram and advantages of a proposed system. The Literature review has been obtained from different source materials including Books, Encyclopedia, Internet surfing and other related articles.

2.1 EXISTING SYSTEM

The existing system of tomato sorting is conducted through;

i. Visual inspection

Involves examining the tomatoes for various attributes such as colour, size, shape, ripeness, and the presence of defects. And determining whether to store as a fresh tomato, take to immediate use or dump if rotten.

ii. Handpicking

Involves manually selecting and picking tomatoes from the plants based on their ripeness and quality.

Workers manually inspect each tomato, sorting them based on size, color, and visible defects. This method relies heavily on the experience and judgment of the workers. Unlike machine human judgment varies as well as they have exhausted energy that is not constant from when the work starts to when it ends, these may lead to inconsistency of sorting quality.

The existing system consists of three storage location, whereby one is used to store fresh tomatoes that are not exactly ripe and must be green in color which are later taken to the pantry for storage, in the pantry the tomatoes are sprayed with anti-pest formulas to prevent pests from destroying the tomatoes while there in storage. The other storage location is for tomatoes that are already ripe and ready to be used immediately, these can be taken to the market place instantly after they have been harvested. The last storage location is for storing the rotten tomatoes or tomatoes that were destroyed by insects, birds or pesticides; they can be dried on direct sunlight so as to obtain tomato seeds that can be used in the next plantation.

2.1 A BLOCK DIAGRAM OF EXISTING SYSTEM

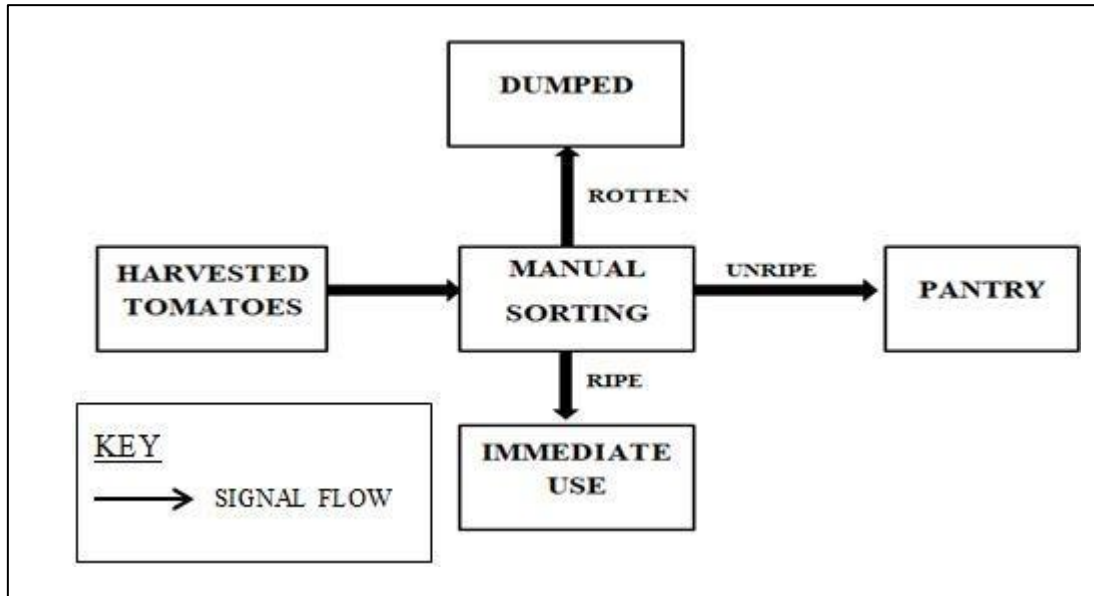


Figure 2.1: block diagram of existing system

2.1.1 ADVANTAGES OF EXISTING SYSTEM

- i. **Adaptation to various quality standards in sorting.** Human workers can adapt to different sorting criteria and handle a wide variety of produce with varying characteristics. This flexibility is particularly useful when dealing with different tomato varieties or changing market demands.
- ii. **Quick corrective action.** Human workers can make subjective assessments based on their experience and intuition, which can be valuable in ensuring the quality and marketability of the tomatoes.
- iii. **Ideal for small scale operation.** For small-scale operations, manual sorting can be more practical and efficient than investing in automated systems.
- iv. **Adaptable for various environments.** It is easy to modify the sorting process based on changes in the product, criteria, or market requirements. For example, a shift in customer preferences may lead to the need for more stringent quality checks, and manual sorting can adapt to such changes without requiring a system overhaul.

2.1.2 DISADVANTAGES OF EXISTING SYSTEM

- i. **Inconsistence and poor judgment.** Human workers can make mistakes, such as misclassifying items or overlooking defects. Unlike machines that follow predefined criteria, humans can sometimes introduce inconsistencies in sorting, leading to variations in product quality.
- ii. **A time consuming activity and needs large manual labors,** which may lead to higher operating cost as well as time consuming for a small load. Manual sorting is typically much slower than automated systems, especially for large volumes of items. Sorting large quantities of tomatoes or other produce by hand can result in bottlenecks and longer processing times, limiting overall throughput.
- iii. **Easy spreading of the diseases among the workers,** there is a high chance of contamination. Manual sorting often involves repetitive motions, which can lead to injuries or musculoskeletal disorders such as carpal tunnel syndrome, back pain, or eye strain from long hours of close inspection.
- iv. **Scaling up operations with manual sorting can be challenging,** as it requires more workers and space. This can limit the ability to handle larger volumes efficiently. Manual sorting becomes increasingly less efficient as the volume of items to be sorted grows. As production needs scale up, manual sorting can struggle to keep up, making it difficult to meet the demands of large-scale operations or rapidly growing businesses.
- v. New workers need to be trained on sorting standards and techniques, which can be time-consuming and costly.
- vi. **Challenges in Detecting Minor Defects.** While skilled workers can identify defects, they may struggle to detect very small or subtle issues, such as minor bruises, internal damage, or slight color variations. Automated systems, on the other hand, can be programmed to detect even the smallest irregularities with higher precision.
- vii. **Scheduling and Labor Shortages.** Managing a large workforce for manual sorting can be complex. Ensuring there are enough workers available during peak seasons, handling shift rotations, and addressing absences can become a logistical challenge.

2.2 PROPOSED SYSTEM

A proposed system introduces a new way of sorting tomatoes that integrates machine learning with post-agricultural processes, promising to eliminate setbacks of the manual sorting system.

The system will include a built-in instructions synchronized with sets of input and output through a central processing system (MCU).

The system will have a defined quality standard of sorting that will ensure a higher efficiency, reduces human error, and increases the speed of processing, making it ideal for large-scale farms.

The system will detect the color of tomatoes to determine their ripeness and sort them accordingly, then measure the dimensions of tomatoes to ensure uniformity in size.

For the robotic arm will be used to pick and place tomatoes into different categories based on their quality either unripe, ripe or rotten.

The system will use a user-friendly interface that allows operators to monitor and control the sorting process easily. The machine learning algorithms used in making sorting decisions will be based on patterns learned from data. It can continually improve its sorting accuracy as it processes more data over time.

The system can be adjusted to sort tomatoes according to different criteria such as ripeness, size, weight, and external appearance. Users can set up sorting rules to meet specific market or customer requirements. Also intelligent sorting systems will be designed to handle large volumes of tomatoes quickly. High-speed sorting enables more efficient processing in large-scale operations, reducing labor and increasing throughput.

Compared to human being this system will be able to run continuously, 24/7, without requiring frequent breaks or rest periods, or being tired making them ideal for high-demand agricultural operations.

By automating the sorting process, human errors such as misclassification, inconsistent grading, or fatigue-induced mistakes are minimized, leading to more reliable outcomes. With less reliance on human labor, labor costs are significantly reduced, especially in larger operations that would require a substantial workforce for manual sorting.

2.2 BLOCK DIAGRAM OF PROPOSED SYSTEM

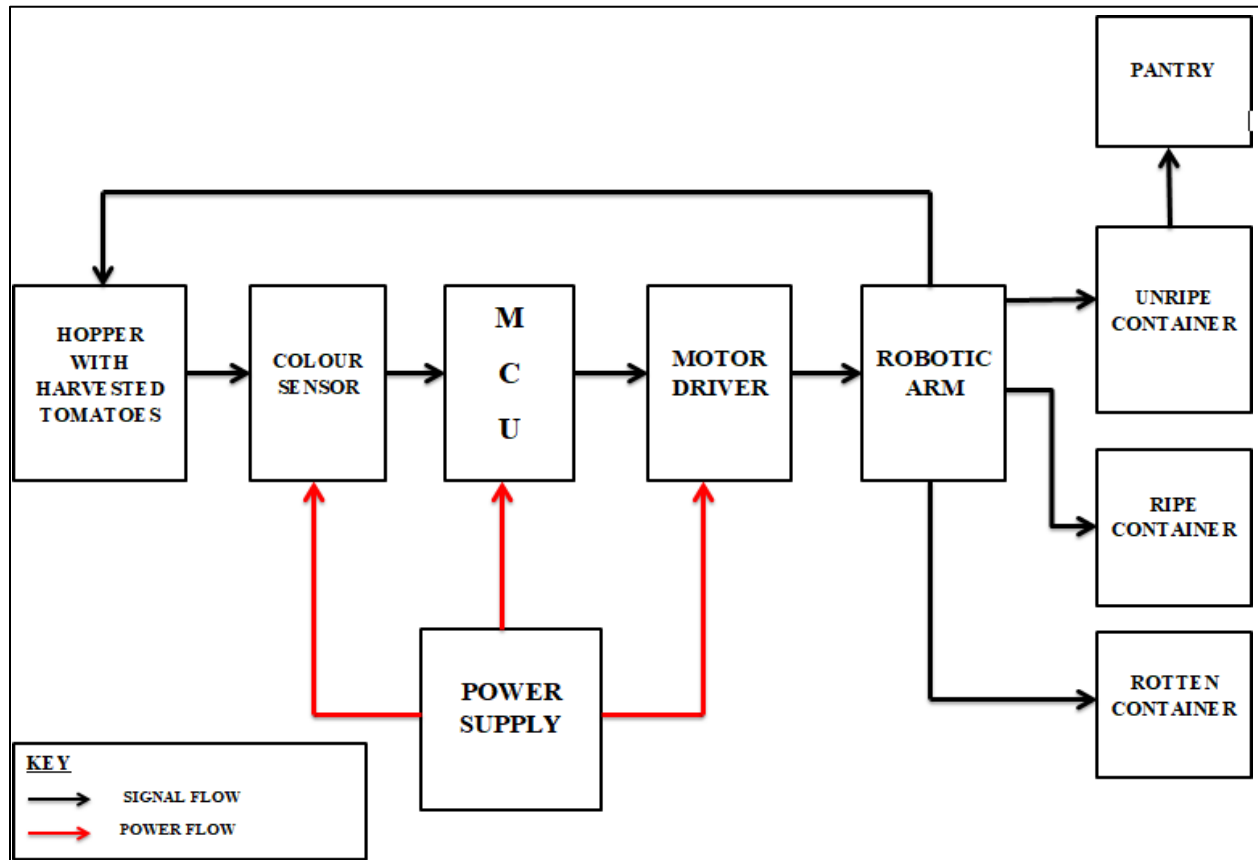


Figure 2.2: block diagram of proposed system

2.2.1 OPERATION OF THE PROPOSED SYSTEM

The system is powered by 220VAC which is rectified up to 12Vdc to be used by the system components. The system is made up of set of inputs and output linked by a microcontroller which acts as a central processing of all actions.

Set of input is made up of a color sensor together with a hopper, for a color sensor, it identifies the color of tomato at the moment and generate an appropriate voltage that will act as input signal to the microcontroller.

Tomatoes are fed into the system through a hopper. This serves as the entry point where harvested tomatoes are collected and prepared for sorting.

The tomatoes are passed through a color sensor, which detects their color. The sensor is programmed to identify the ripeness level based on predefined color thresholds (e.g., green for unripe, red for ripe or darker colors for rotten).

The detected color information is sent to the MCU, which processes the input data and decides the sorting action based on the identified category of the tomato (unripe, ripe, or rotten). MCU allows operators to monitor and control the sorting process, adjust settings, and view real-time data.

Based on the decision from the MCU, the motor driver controls the movement of the robotic arm. It provides the necessary electrical signals to the robotic arm to execute sorting actions.

The robotic arm physically moves the tomatoes to the respective containers. If

- i.** If the tomato is unripe (detected as green), it is placed in the Unripe Container.
- ii.** If the tomato is ripe (detected as red), it is placed in the Ripe Container:
- iii.** If the tomato is identified as rotten, it is moved to the Rotten Container

Tomatoes in the ripe container may eventually be directed to the pantry for storage or further processing.

Set of output is made up of motor driver coupled with a robotic arm. After microcontroller processes the received input signal from the color sensor, it generates an output signal feeding it to the motor driver that will control the motion of coupled motors in a robotic arm, so that the arm may move with accordance to the generated signal. A robotic arm after picking up tomato it takes it to the appropriate storage container.

Throughout the operation, the system continuously monitors the sorting process, capturing data on throughput; reject rates, and sorting accuracy. This allows for immediate intervention if any issues arise (e.g., system malfunctions or unusually high rejects rates). The system allows operators to adjust settings remotely based on real-time analysis. For example, if a batch of tomatoes is unusually large or small, the sorting parameters can be quickly modified to account for the change.

The system continuously monitors the sorting process, providing real-time feedback on the quality of sorted tomatoes. Any deviations from the desired quality standards can be quickly identified and corrected.

For Lower-quality tomatoes or those not suitable for fresh consumption are sent to immediate use. While rotten ones are discarded.

2.2.2 DESCRIPTION OF COMPONENTS OF PROPOSED SYSTEM

2.2.2.1 COLOR SENSOR

This is a type of sensor used to detect and measure the color of an object or surface. It typically works by analyzing the light reflected from an object, breaking it down into the components of the visible light spectrum (red, green, blue, etc.), and then determining the color based on the proportions of these components.

In the project of an intelligent tomato sorting system, a color sensor helps identify the ripeness and quality of tomatoes by analyzing their color. The color of a tomato can be an important indicator of its ripeness, quality, and suitability for different purposes (e.g., fresh sale or processing).

Basically, color sensor will be used to detect the color of tomato by emitting light towards the tomato and detect the color of reflected light from tomato, and then they can detect the intensity of light reflected from tomato and makes comparison and differentiating with the primary colors like red, blue, and green. These are also called as color detectors.

BASIC OPERATION OF COLOR SENSOR

Color sensors generally consist of the following key components namely;

i. Light Source

The sensor includes a light source, often an LED, that emits light onto the object (in this case, a tomato). This light is typically in the visible spectrum (from violet to red), but some sensors may use additional wavelengths (infrared or ultraviolet) for more specific applications.

ii. Optical Filter

After the light hits the object, the sensor uses optical filters to separate the reflected light into different color components (red, green, blue, or RGB). This filtering allows the sensor to analyze each color individually.

iii. Photo detector

The reflected light is then detected by a photo detector or a photodiode array, which converts the light into an electrical signal. This signal is proportional to the intensity of the reflected light in each of the color channels (red, green, blue).

iv. Color Processing Unit

The electrical signals generated by the photo detector are sent to a processing unit (often a microcontroller or processor). This unit converts the signals into color values, typically represented in the RGB (Red, Green, Blue) format, or another color space like HSV (Hue, Saturation, Value), depending on the sensor and application.

v. Analog-to-Digital Converter (ADC)

The analog signals from the photodiodes are converted into digital signals using an ADC. This allows the sensor to process and analyze the data.

vi. Output

The processed color data is used by the system to make decisions. For example, it can determine if a tomato is ripe (based on its color), or sort tomatoes into different categories (e.g., red, green, yellow).

Color sensors help determine the ripeness of tomatoes by detecting the color changes as they mature. For example, tomatoes transition from green to yellow and then to red as they ripen. Sorting based on these color changes allows the system to classify tomatoes as under-ripe, ripe, or over-ripe.

Tomatoes with uneven color or discoloration can be flagged as defective or overripe. Color sensors can identify these imperfections, ensuring that only the highest-quality tomatoes are sorted for storage.

Unlike human inspectors, color sensors provide consistent and precise color analysis, ensuring that the sorting process remains uniform and unbiased over time.



Figure 2.3: diagram of color sensor

2.2.2.2 MOTOR DRIVER

A motor driver is an electronic device that controls the operation of a motor. It acts as an interface between a microcontroller (or other control systems) and the motor, allowing the microcontroller to control the motor's speed, direction, and torque.

A motor driver is a module that works by controlling both the speed and rotation direction of motors. This module uses two techniques for the control speed and rotation direction of the motors. These are PWM – For controlling the speed and H-Bridge – For controlling rotation direction.

BASIC OPERATION OF MOTOR DRIVER

Following are key features of the motor driver;

i. Input Signals

A motor driver receives control signals from a microcontroller or other control systems. The signals are used to determine;

- **Direction**

Whether the motor should rotate clockwise or counterclockwise

- **Speed**

The speed at which the motor should run, usually controlled by adjusting the voltage or using techniques like Pulse Width Modulation (PWM). Motor drivers often use PWM signals to control the speed of the motor. By varying the duty cycle of the PWM signal, the driver can adjust the average voltage supplied to the motor, thereby controlling its speed.

- **Enable/Disable**

Whether the motor should be running or stopped.

ii. Power Amplification

Motor drivers amplify the low-power control signals from the microcontroller to provide the necessary current and voltage to drive the motor. They often include protection features such as overcurrent protection, thermal shutdown, and short-circuit protection to prevent damage to the motor and the driver.

iii. Direction Control

Many motor drivers use an H-bridge circuit to control the direction of the motor. By changing the polarity of the voltage applied to the motor, the H-bridge allows the motor to rotate in both clockwise and counterclockwise directions.

iv. Interface with Microcontroller

- **Control Pins**

Motor drivers have control pins that connect to the microcontroller. These pins receive control signals (such as PWM signals) from the microcontroller to regulate the motor's operation.

- **Feedback**

Some motor drivers provide feedback signals to the microcontroller, such as current sensing or fault detection, to monitor the motor's performance and ensure safe operation.

v. Output

The motor driver then outputs the amplified signal to the motor's terminals, which powers the motor. The motor driver ensures that the motor receives stable and consistent power to prevent damage and improve efficiency.

Motor driver that will be used in completion of this project is a DC motor drivers control the direction and speed of a DC motor. A common motor driver for DC motors is the L298N, which allows for bidirectional control (clockwise or counterclockwise) and speed control using PWM.

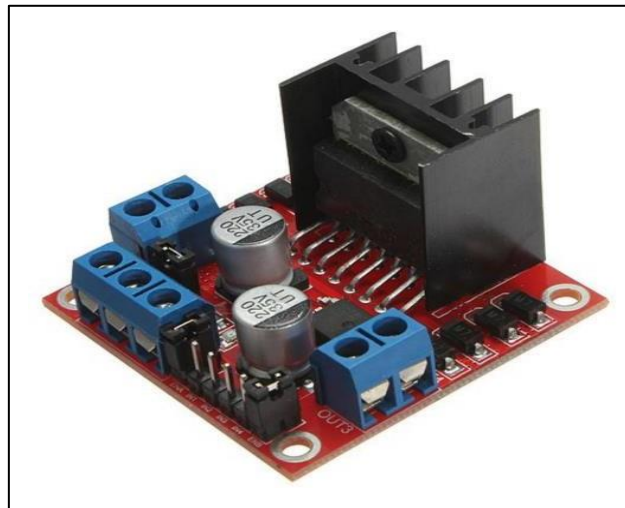


Figure 2.4: diagram of motor driver

2.2.2.3 SERVO MOTOR

This is a type of motor that is designed for precise control of angular or linear position, velocity, and acceleration. It is commonly used in applications that require accurate and controlled movement, such as robotics, automation, and CNC machinery.

Unlike regular DC motors, which rotate continuously, servo motors are typically used in applications that require controlled rotation over a limited range, usually within 0 to 180 degrees (or even more, depending on the motor).

This is an electric motor that provides control of angular or linear position, speed and torque based on an input signal from a controller.

This will provide a robotic arm with joint control, torque generation, speed management as well as necessary movements.

BASIC OPERATION OF SERVO MOTOR

Operation of servo motor is divided into the following categories;

i. Command Signal

The control system sends a command signal to the servo motor, specifying the desired position, speed, or torque. The servo motor receives an input signal, typically a Pulse Width Modulation (PWM) signal, which indicates the desired position of the motor shaft. The duration (width) of the pulse determines how far the servo should rotate.

ii. Feedback

The feedback device (encoder or potentiometer) continuously monitors the motor's position and sends this information back to the control circuit. The potentiometer inside the motor continuously tracks the position of the motor shaft. The feedback from the potentiometer is compared to the desired position (set by the PWM signal).

iii. Error Calculation

The control circuit compares the actual position of the motor with the desired position and calculates the error. The feedback loop ensures the motor stops at the exact desired position, maintaining precise control over movement. Once the servo reaches the target position, it holds that position without needing constant input signals.

iv. Adjustment

Based on the error, the control circuit adjusts the motor's operation to minimize the error and achieve the desired position. This process is repeated continuously to maintain precise control. If the motor's current position does not match the desired position, the internal control circuitry adjusts the power supplied to the motor, causing it to rotate in the correct direction until it reaches the desired position.

Here are some key features and components of a servo motor:

i. DC Motor

The core of a servo motor is usually a small DC motor that provides continuous rotational motion. This motor's rotation is controlled and limited to a certain angle by the servo mechanism.

ii. Feedback Device (Potentiometer)

A potentiometer or encoder is used in the servo motor to provide feedback to the control system. It continuously monitors the position of the motor shaft and ensures that the motor is turning to the desired position.

iii. Control Circuitry

The motor is controlled by an electronic circuit that receives input signals and adjusts the motor's rotation accordingly. The control circuitry interprets the feedback from the potentiometer and makes adjustments to maintain the desired position.

iv. Gearbox

The motor is usually connected to a set of gears that reduce the speed of the motor's rotation while increasing its torque. This ensures that the servo motor has sufficient power to move objects or systems in a controlled manner. Many servo motors include a gearbox to reduce the speed and increase the torque of the motor. This allows for finer control of movement.

Servo motors are essential components in various applications that require precise and controlled movement. Their ability to provide accurate positioning, variable speed, and high torque makes them ideal for a wide range of industrial and consumer applications.



Figure 2.5: diagram of Servo motor

2.2.2.4 MICROCONTROLLER

This is a compact integrated circuit designed to govern a specific operation in an embedded system. It includes a processor, memory, and input/output (I/O) peripherals on a single chip. It is a small, self-contained computer on a single chip. It is designed to perform specific tasks and control systems in embedded applications.

Microcontrollers are widely used in electronics and embedded systems where they control devices like home appliances, robots, automotive systems, medical devices, and consumer electronics.

It is a small, integrated computer system that contains a processor, memory, and input/output (I/O) peripherals on a single chip. This is the heart of the whole system. It controls most of the operations in a system. It receives set of input signals from the sensor processes the received signal and generate an output signal that if fed to the set of output devices.

BASIC OPERATION OF MICROCONTROLLER

i. Program Execution

The microcontroller executes a program stored in its memory. This program consists of a series of instructions that control the operation of the microcontroller and its peripherals.

ii. Input Processing

The microcontroller reads input signals from sensors and other devices through its I/O pins. It processes these inputs based on the program instructions.

iii. Processing

The microcontroller processes the input data according to the logic defined in its program. For example, it may read data from a sensor, process it, and make decisions based on certain conditions. These inputs can come from sensors, switches, or communication interfaces.

iv. Output Control

Based on the processed inputs and program logic, the microcontroller generates output signals to control actuators, displays, and other devices connected to its I/O pins. This could involve turning on a motor, displaying information on an LCD screen, or sending data to another device.

v. Communication

The microcontroller communicates with other devices and systems through its communication interfaces. This allows it to exchange data and coordinate actions with external components.

Key Components of a Microcontroller

i. Central Processing Unit (CPU)

The brain of the microcontroller, responsible for executing instructions and performing calculations. It processes data and controls the operation of the microcontroller. It handles the logic and control of the microcontroller's operations.

ii. Memory

RAM (Random Access Memory): Temporarily stores data while the microcontroller is running programs. It is used for holding variables, buffers, and temporary data.

ROM (Read-Only Memory): Stores the microcontroller's firmware (the program that tells the microcontroller what to do). Flash memory is often used for ROM in modern microcontrollers as it can be reprogrammed.

EEPROM (Electrically Erasable Programmable Read-Only Memory): A type of non-volatile memory used to store small amounts of data that need to persist even when the device is powered off.

iii. Input/output (I/O) Peripherals

These pins allow the microcontroller to interact with external devices like sensors, motors, and displays. Some pins are digital (for high or low signals), while others may be analog (to read continuous signals, like temperature or light levels).

Digital I/O Pins: Used to interface with digital devices such as switches, LEDs, and sensors. These pins can be configured as either input or output.

Analog I/O Pins: Used to interface with analog devices such as sensors and actuators. These pins typically include analog-to-digital converters (ADCs) and digital-to-analog converters (DACs).

iv. Timers and Counters

Microcontrollers often include built-in timers and counters that can be used to measure time, generate delays, or produce periodic events. These are especially important for applications that require time-based control, such as PWM (Pulse Width Modulation) signals or clock-driven systems.

v. Communication Interfaces

Microcontrollers come with communication protocols such as SPI (Serial Peripheral Interface), I2C (Inter-Integrated Circuit), UART (Universal Asynchronous Receiver-Transmitter), or USB that allow them to communicate with other devices like sensors, other microcontrollers, or computers.

vi. ADC/DAC (Analog-to-Digital Converter/Digital-to-Analog Converter)

ADCs allow the microcontroller to read analog signals (such as sensor outputs) and convert them into digital values that it can process. DACs, on the other hand, allow the microcontroller to output analog signals.

vii. Oscillator/Clock

Microcontrollers typically have an internal clock or an external crystal oscillator that provides the timing reference to coordinate operations and execute instructions at a steady pace.

viii. Power Supply

Provides the necessary voltage and current to operate the microcontroller. Some microcontrollers include power-saving features to reduce energy consumption.



Figure 2.6: diagram of microcontroller

2.2.2.5 POWER SUPPLY

This is an electrical device that provides the necessary electrical power to operate electronic devices and systems. It converts electrical energy from a source, such as a wall outlet or battery, into the appropriate voltage, current, and frequency required by the connected devices

This is an essential part of the system as it serves to supply power to all components in a circuit. A system will be powered by 220 V.A.C, a step-down transformer is used to step down voltage to 12V then a rectifier is used to convert AC to DC suitable for the operation of electronic components including microcontroller.

BASIC OPERATION OF POWER SUPPLY

i. Voltage and current Regulation

Power supplies provide a stable and consistent output voltage, ensuring that connected devices receive the correct voltage for proper operation. Power supplies include current limiting features to prevent excessive current flow, protecting connected devices from damage.

Some power supplies can operate in constant current mode, providing a steady current output regardless of changes in load resistance. Some power supplies offer adjustable output voltage, allowing users to set the desired voltage level for specific applications.

ii. Energy Conversion

Power supplies are designed to efficiently convert electrical energy from the source to the output, minimizing energy loss and heat generation.

iii. Ripple and Noise Reduction

Power supplies, especially in sensitive applications, work to reduce unwanted variations in voltage (called ripple) and electrical interference (called noise) that might affect the performance of the powered device.

iv. Protection Features

Overvoltage Protection: Prevents the output voltage from exceeding a safe level, protecting connected devices from damage.

Overcurrent Protection: Limits the current flow to prevent excessive current that could damage the power supply or connected devices.

Short-Circuit Protection: Detects and responds to short-circuit conditions, preventing damage to the power supply and connected devices.

Thermal Protection: Monitors the temperature of the power supply and shuts it down if it overheats, preventing damage from excessive heat.

Components of a Power Supply

i. Transformer

Steps up or steps down the input voltage to the desired level. Transformers are commonly used in AC power supplies.

ii. Rectifier

Converts alternating current (AC) to direct current (DC) using diodes or other rectifying components.

iii. Filter

Reduces voltage ripple and smooths the output voltage using capacitors and inductors.

iv. Regulator

Maintains a stable output voltage despite variations in input voltage or load conditions. Regulators can be linear or switching types.

v. Control Circuit

A feedback system that monitors the output and adjusts the power supply's operation to maintain a stable voltage and current, ensuring reliable operation.

vi. Heat Sink

Dissipates heat generated by the power supply components, preventing overheating and ensuring reliable operation.

Different types of power supply

i. Linear Power Supplies

Commonly used in audio equipment, laboratory instruments, and low-noise applications.

ii. Switching Power Supplies

Use high-frequency switching regulators to convert electrical energy. They are more efficient than linear power supplies and can provide higher power output.

iii. Uninterruptible Power Supplies (UPS)

Provide backup power in case of a power outage. They include batteries that supply power when the main power source is unavailable.

iv. Battery Power Supplies

Use batteries as the primary power source. They provide portable and reliable power for various devices.

2.2.2.5 DIAGRAM OF POWER SUPPLY

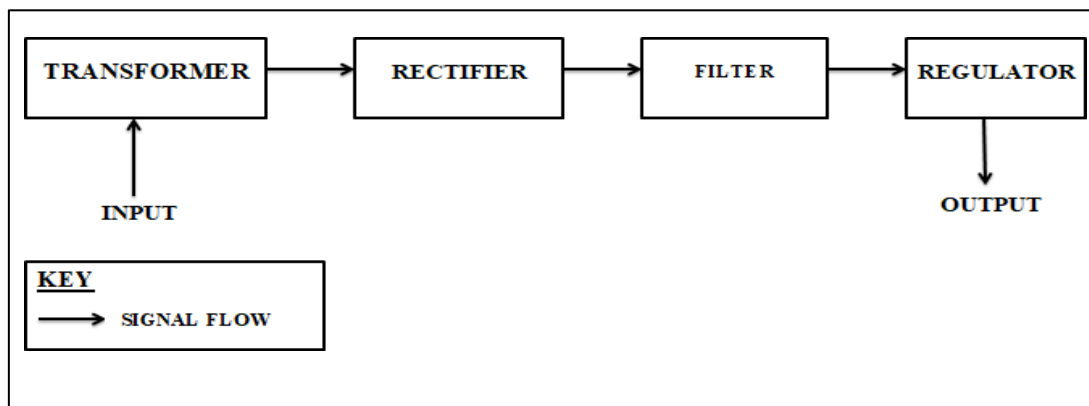


Figure 2.7: diagram of power supply

2.2.3 ADVANTAGES OF PROPOSED SYSTEM

- i. This system eliminates human errors caused by fatigue or inconsistent judgment, improving the overall quality of the sorted tomatoes. The project will have ability to eliminate human errors caused by fatigue or inconsistent judgment, thereby improving the overall quality of the sorted tomatoes
- ii. The system can sort tomatoes much faster than human labor, significantly increasing throughput and reducing the time required for sorting, also operate continuously without breaks enhancing productivity.
- iii. The intelligent sorting can help minimize the number of tomatoes that are incorrectly categorized or discarded due to human mistakes.
- iv. Intelligent sorting systems can operate continuously, sorting items much faster than human workers. This speed significantly increases overall throughput and productivity, especially in high-volume industries.
- v. These systems can work around the clock without the need for breaks or downtime, allowing for continuous production and processing, which improves operational efficiency.
- vi. By automating the sorting process, businesses can reduce the need for manual labor, which can be expensive and inefficient over time. Although the initial investment in technology may be high, the long-term savings on labor costs and human resources make intelligent sorting systems cost-effective.
- vii. Automated sorting minimizes mistakes and misclassification, which helps in reducing product waste, minimizing the need for rework, and improving the yield of quality products.

CHAPTER THREE

3.0 DATA COLLECTION

This chapter concerns with gathering necessary data that will facilitate in the accomplishment of the project. Relevant data concerning the project has been collected through Literature review which involved reading from different books, journals and surfing from the internet. Also, the collection of data has been conducted through consulting different professionals and experts related to the project at Ulanga pantry and site visiting hence provision of the key parameters of the existing system that guides in analysis and actual implementation of the proposed system.

The collected Data is divided into two categories which are **Primary data** and **Secondary data** and are explained as follows; -

3.1 PRIMARY DATA

The preliminary information regarding the project was obtained through the collection of data. This was through observation and then after, the interview and questionnaires with appropriate professionals concerning tomato sorting system was conducted. Clear information regarding the existing system together with the challenges facing it, were given.

The table below shows the different categories of tomatoes

TOMATOES	COLOR	CONDITION	DAYS TAKEN TO RIPE WHEN PRESERVED
Unripe	Green	Firm and hard	0 – 10 days
Half ripe	Partial red or orange	Starts to soften	0 – 4 days
Ripe	Red	Softened	Zero
Unripe + ripe	Green and red	Hard mixed with softened tomatoes	0 – 6 days

Table 3.1: Categories of tomatoes

3.1.1 OBSERVATIONS

Through observation of the existing system, the following points were noted.

- i. The system is human dependency hence not automated.
- ii. There is high possibility of mixing rotten tomatoes with unripe tomatoes that may lead to fast rotting of tomatoes after being preserved.

3.2 SECONDARY DATA

The information collected under this part are more of technical and are based on the description of devices which are going to be used for the project.

The following are description of main components used in a project;

Table 3.2: Microcontroller specifications

1.	Rated operating voltage	5VDC
2.	I/O digital pins	14
3.	Analog input pins	8
4.	Clock speed	16MHZ
5.	Microcontroller	AT MEGA 328P

Table 3.3: Motor driver

1.	Input voltage	5VDC
2.	Input current	4.5A
3.	PMW frequency	20kHz

Table 3.4: Servo motor

1.	Input voltage	4.8VDC
2.	Torque	2.5kg/cm
3.	Operating speed	0.12s/60°
4.	Rotation	0° – 180°

5.	Duty cycle	5% - 10% at 50Hz
6.	Operating temperature	-30°C – 60°C

Table 3.5: Relay

1	Input voltage	5VDC
2	Input current	0.1 - 2A
3	Turn on time	½ Cycles + 1ms
4	Turn off time	½ Cycles + 1ms

Table 3.6: Color sensor

1.	Input voltage	3.3VDC
2.	Input current	15mA – 30mA
3.	Response time	10ms – 50ms
4.	Color modes	RGB

APPENDICES

Appendix A: Schedule Activities of the Project

ACTIVITIES	Nov 2024	Dec 2024	Jan 2025	Feb 2025	Mar 2025	Apr 2025	May 2025	Jun 2025	Jul 2025
Selection of project title									
Title defending									
Literature review									
Consultation									
Data collection									
Data analysis									
Circuit designing and simulation									
Building and testing prototype									
Report writing									

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